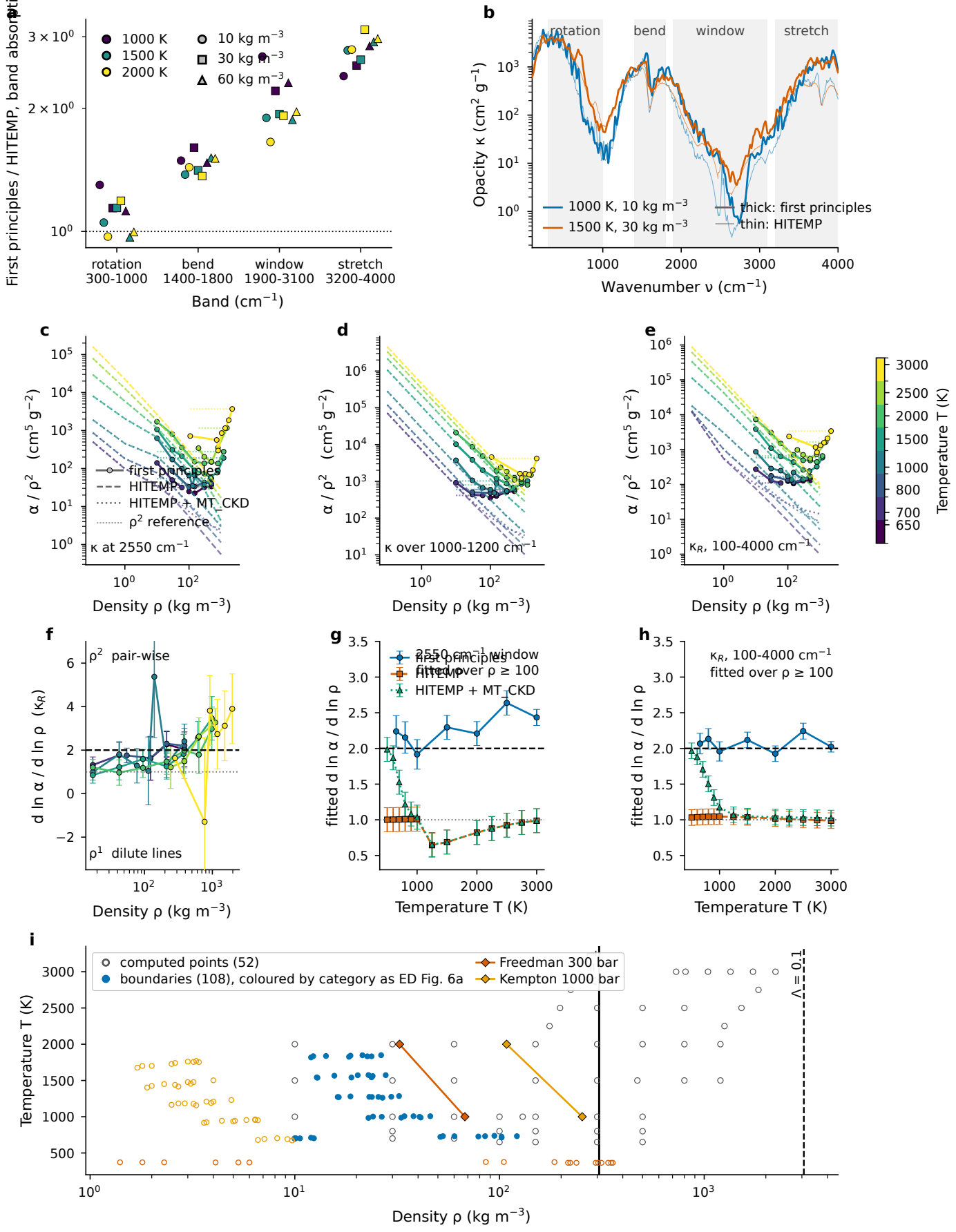

a
First principles / HITEMP, band absorption
1000 K
1500 K
2000 K
10 kg m⁻³
30 kg m⁻³
60 kg m⁻³
rotation 300-1000
bend 1400-1800
window 1900-3100
stretch 3200-4000
Band (cm⁻¹)
b
rotation
bend
window
stretch
Opacity κ (cm² g⁻¹)
1000 K, 10 kg m⁻³
1500 K, 30 kg m⁻³
thick: first principles
thin: HITEMP
Wavenumber ν (cm⁻¹)
c
α / ρ² (cm⁵ g⁻²)
first principles
HITEMP
HITEMP + MT_CKD
ρ² reference
κ at 2550 cm⁻¹
Density ρ (kg m⁻³)
d
κ over 1000-1200 cm⁻¹
e
κR, 100-4000 cm⁻¹
Temperature T (K)
f
d ln α / d ln ρ (κR)
ρ² pair-wise
ρ¹ dilute lines
g
fitted d ln α / d ln ρ
2550 cm⁻¹ window
fitted over ρ ≥ 100
Temperature T (K)
h
κR, 100-4000 cm⁻¹
i
Temperature T (K)
computed points (52)
boundaries (108), coloured by category as ED Fig. 6a
Freedman 300 bar
Kempton 1000 bar
Λ = 0.1